Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Germann Bros Dairy

Physical address of dairy:

1236 E Taylor RD	Ceres	Stanislaus	95307
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X010-X001-XXXX

B. OPERATORS

Germann, Andy or Charlie

Operator name: Germann, Andy or Charlie

Telephone no.: (209) 537-9568 (Landline) (Cellular)

1936 E Taylor RD	Ceres	CA	95307
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Germann, Andy or Charlie

Legal owner name: Germann, Andy or Charlie

Telephone no.: (209) 537-9568 (Landline) (Cellular)

1936 E Taylor RD	Ceres	CA	95307
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	40	40	30	20
Number under roof	150	15	0	0	0	0
Maximum number	160	18	50	45	35	25
Average number	150	16	40	40	30	20
Avg live weight (lbs)	1,400	1,450	950	700		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 5,045.16 *tons per reporting period*

Total nitrogen from manure: 63,374.01 *lbs per reporting period*

After ammonia losses (30% loss applied): 44,361.81 *lbs per reporting period*

Total phosphorus from manure: 10,367.59 *lbs per reporting period*

Total potassium from manure: 28,726.38 *lbs per reporting period*

Total salt from manure: 74,306.70 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,555,000 *gallons*

Total nitrogen generated: 10,307.91 *lbs*

Total phosphorus generated: 2,225.96 *lbs*

Total potassium generated: 14,228.02 *lbs*

Total salt generated: 101,717.21 *lbs*

	3,555,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,555,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW 3	Ground water
IW 5	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/16/2023	Solid commercial fertilizer *UN 32*	2.00 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	1,278.72	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	1,278.72	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Blaker	6	6	2	manure	X041-X032-X003-XXXX
Corner1	40	40	2	manure	X041-X050-X007-XXXX
Dairy1	18	18	2	process wastewater	X022-X001-X001-XXXX
Dairy2	14	14	1	process wastewater	X022-X001-X001-XXXX
Totals for areas that were used for application	78	78	7		
Totals for areas that were not used for application					
Land application area totals	78	78	7		

B. CROPS AND HARVESTS

Blaker

Field name: Blaker

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	68.00 *ton*	As-is		60.2	17,100.00	1,870.00	18,700.00		9.71

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	11.33	387.60	42.39	423.87	875.97

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	154.80 *ton*	As-is		71.5	3,310.00	1,330.00	5,490.00		9.67

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.80	170.80	68.63	283.28	1,422.07

Corner1

Field name: Corner1

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	604.00 *ton*	As-is		70.8	9,840.00	1,230.00	10,500.00		9.24

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.10	297.17	37.15	317.10	814.82

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	1,050.00 *ton*	As-is		72.4	2,950.00	910.00	3,770.00		5.84

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.25	154.88	47.78	197.93	846.22

Dairy1

Field name: Dairy1

09/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 09/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	266.00 *ton*	As-is		77.0	11,500.00	1,060.00	13,600.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.78	339.89	31.33	401.96	693.37

Dairy1

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/05/2023	480.60 *ton*	As-is		74.4	3,510.00	750.00	5,400.00		6.96

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.70	187.43	40.05	288.36	951.46

Dairy2

Field name: Dairy2

03/02/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 14 Plant date: 03/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/02/2023	566.00 *ton*	As-is		75.6	6,880.00	650.00	7,520.00		15.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	1,800.00	162.00	1,260.00	0.00
Total actual harvest content	40.43	556.30	52.56	608.05	3,038.29

NUTRIENT BUDGET

A. LAND APPLICATIONS

Blaker - 10/31/2022: Oats, silage-soft dough

Field name: Blaker

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	284.08	191.86	864.96	6,105.60	150.00 *ton*
Application event totals		284.08	191.86	864.96	6,105.60	

Blaker - 05/27/2023: Corn, silage

Field name: Blaker

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	30.34	807,840.00 *gal*
Application event totals		0.00	0.00	0.00	30.34	

05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	203.08	38.86	108.41	4,134.63	90.00 *ton*
Application event totals		203.08	38.86	108.41	4,134.63	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	30.34	807,840.00 *gal*
Application event totals		0.00	0.00	0.00	30.34	

Blaker - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	30.34	807,840.00 *gal*
Application event totals		0.00	0.00	0.00	30.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	30.34	807,840.00 *gal*
Application event totals		0.00	0.00	0.00	30.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	30.34	807,840.00 *gal*
Application event totals		0.00	0.00	0.00	30.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	25.28	673,200.00 *gal*
Application event totals		0.00	0.00	0.00	25.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	25.28	673,200.00 *gal*
Application event totals		0.00	0.00	0.00	25.28	

Corner1 - 10/30/2022: Oats, silage-soft dough

Field name: Corner1

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Corner1 - 10/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	284.08	191.86	864.96	6,105.60	1,000.00 *ton*
Application event totals		284.08	191.86	864.96	6,105.60	

Corner1 - 05/27/2023: Corn, silage

Field name: Corner1

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	31.85	5,654,880.00 *gal*
Application event totals		0.00	0.00	0.00	31.85	

05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	162.46	31.09	86.72	3,307.70	480.00 *ton*
Application event totals		162.46	31.09	86.72	3,307.70	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	28.82	5,116,320.00 *gal*
Application event totals		0.00	0.00	0.00	28.82	

06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Corner1 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	28.82	5,116,320.00 *gal*
Application event totals		0.00	0.00	0.00	28.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	28.82	5,116,320.00 *gal*
Application event totals		0.00	0.00	0.00	28.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	28.82	5,116,320.00 *gal*
Application event totals		0.00	0.00	0.00	28.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	27.30	4,847,040.00 *gal*
Application event totals		0.00	0.00	0.00	27.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	25.79	4,577,760.00 *gal*
Application event totals		0.00	0.00	0.00	25.79	

Dairy1 - 09/24/2022: Oats, silage-soft dough

Field name: Dairy1

Crop: Oats, silage-soft dough

Plant date: 09/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Dairy1 - 09/24/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.79	12.57	113.07	678.73	300,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	33.71	2,692,800.00 *gal*
Application event totals		65.79	12.57	113.07	712.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.99	23.89	214.84	1,289.58	570,000.00 *gal*
Application event totals		124.99	23.89	214.84	1,289.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.73	18.66	120.56	930.19	660,000.00 *gal*
Application event totals		103.73	18.66	120.56	930.19	

Dairy1 - 05/27/2023: Corn, silage

Field name: Dairy1

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.15	8.48	54.80	422.81	300,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	33.71	2,692,800.00 *gal*
Application event totals		47.15	8.48	54.80	456.52	

Dairy1 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	26.97	2,154,240.00 *gal*
Application event totals		0.00	0.00	0.00	26.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.43	7.64	49.32	380.53	270,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	30.34	2,423,520.00 *gal*
Application event totals		42.43	7.64	49.32	410.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	26.97	2,154,240.00 *gal*
Application event totals		0.00	0.00	0.00	26.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.55	10.41	42.43	355.50	270,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	30.34	2,423,520.00 *gal*
Application event totals		32.55	10.41	42.43	385.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	23.59	1,884,960.00 *gal*
Application event totals		0.00	0.00	0.00	23.59	

Dairy1 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.55	10.41	42.43	355.50	270,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	30.34	2,423,520.00 *gal*
Application event totals		32.55	10.41	42.43	385.83	

Dairy2 - 03/02/2023: Alfalfa, hay

Field name: Dairy2

Crop: Alfalfa, hay

Plant date: 03/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.37	6.54	42.27	326.17	180,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	26.00	1,615,680.00 *gal*
Application event totals		36.37	6.54	42.27	352.17	

05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	28.16	1,750,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.16	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.37	6.54	42.27	326.17	180,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	26.00	1,615,680.00 *gal*
Application event totals		36.37	6.54	42.27	352.17	

Dairy2 - 03/02/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.22	9.67	39.40	330.10	195,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	28.17	1,750,320.00 *gal*
Application event totals		30.22	9.67	39.40	358.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	28.17	1,750,320.00 *gal*
Application event totals		0.00	0.00	0.00	28.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.90	8.93	36.37	304.71	180,000.00 *gal*
IW 3	Ground water	0.00	0.00	0.00	26.00	1,615,680.00 *gal*
Application event totals		27.90	8.93	36.37	330.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	28.17	1,750,320.00 *gal*
Application event totals		0.00	0.00	0.00	28.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.90	8.93	36.37	304.71	180,000.00 *gal*
Application event totals		27.90	8.93	36.37	304.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	26.00	1,615,680.00 *gal*
Application event totals		0.00	0.00	0.00	26.00	

Dairy2 - 03/02/2023: Alfalfa, hay

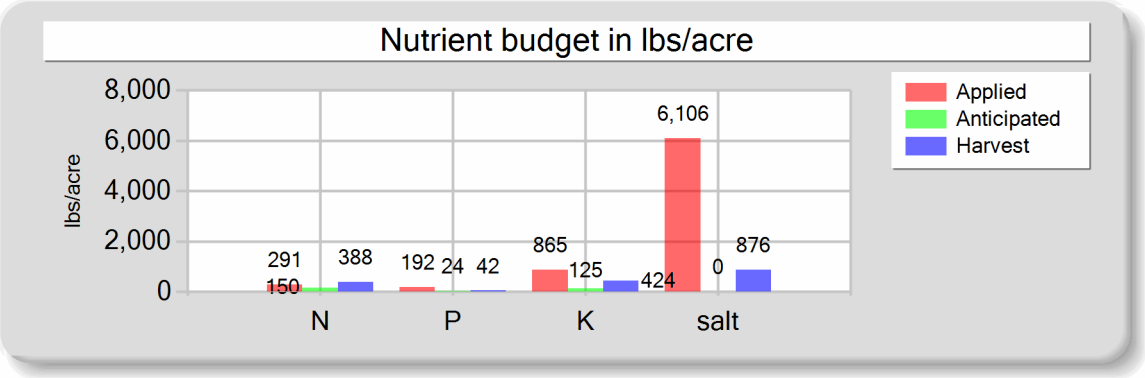
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	26.00	1,615,680.00 *gal*
Application event totals		0.00	0.00	0.00	26.00	

B. NUTRIENT BUDGET

Blaker - 10/31/2022: Oats, silage-soft dough

Field name: Blaker | Crop: Oats, silage-soft dough | Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	284.08	191.86	864.96	6,105.60
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.08	191.86	864.96	6,105.60
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	387.60	42.39	423.87	875.97
Nutrient balance	-96.52	149.47	441.09	5,229.63
Applied to removed ratio	0.75	4.53	2.04	6.97

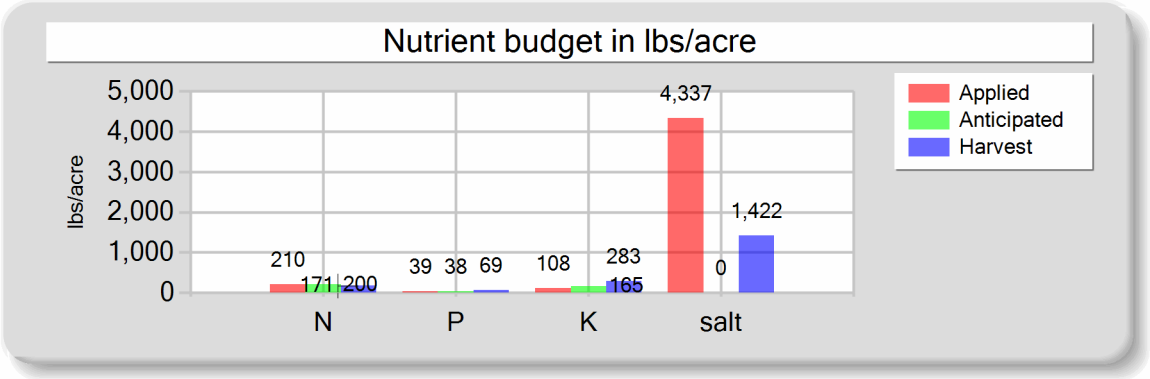
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Blaker - 05/27/2023: Corn, silage

Field name: Blaker | Crop: Corn, silage | Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	203.08	38.86	108.41	4,134.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	202.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.08	38.86	108.41	4,336.87
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	170.80	68.63	283.28	1,422.07
Nutrient balance	39.28	-29.77	-174.88	2,914.80
Applied to removed ratio	1.23	0.57	0.38	3.05

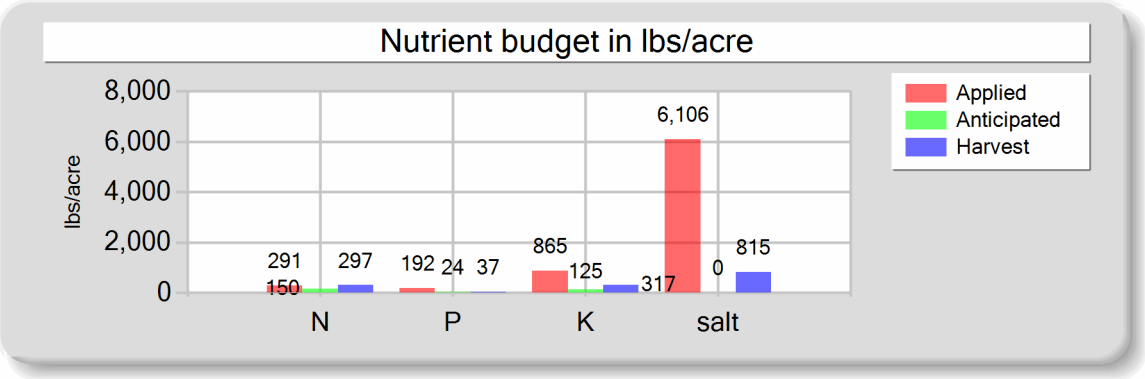
Fresh water applied	
5,385,600.00	*gallons*
198.33	*acre-inches*
33.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Corner1 - 10/30/2022: Oats, silage-soft dough

Field name: Corner1 Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	284.08	191.86	864.96	6,105.60
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.08	191.86	864.96	6,105.60
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	297.17	37.15	317.10	814.82
Nutrient balance	-6.09	154.71	547.86	5,290.78
Applied to removed ratio	0.98	5.17	2.73	7.49

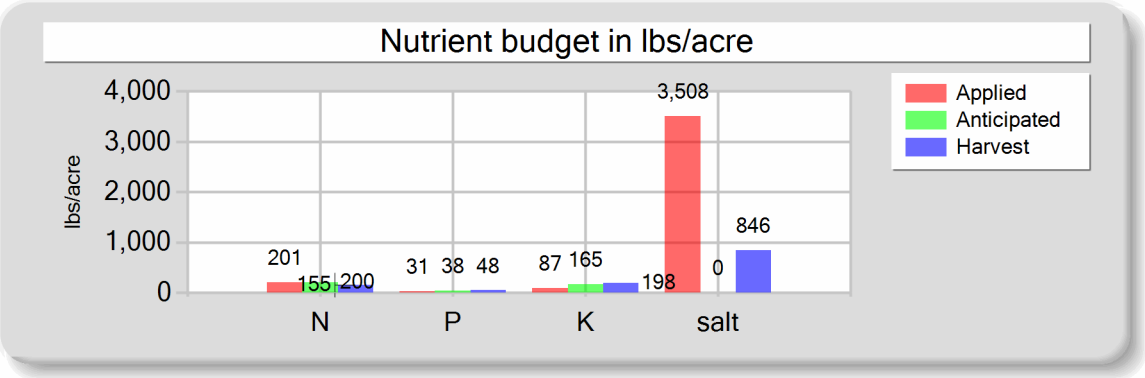
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Corner1 - 05/27/2023: Corn, silage

Field name: Corner1 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	162.46	31.09	86.72	3,307.70
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	200.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.46	31.09	86.72	3,507.92
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	154.88	47.78	197.93	846.22
Nutrient balance	46.59	-16.68	-111.20	2,661.71
Applied to removed ratio	1.30	0.65	0.44	4.15

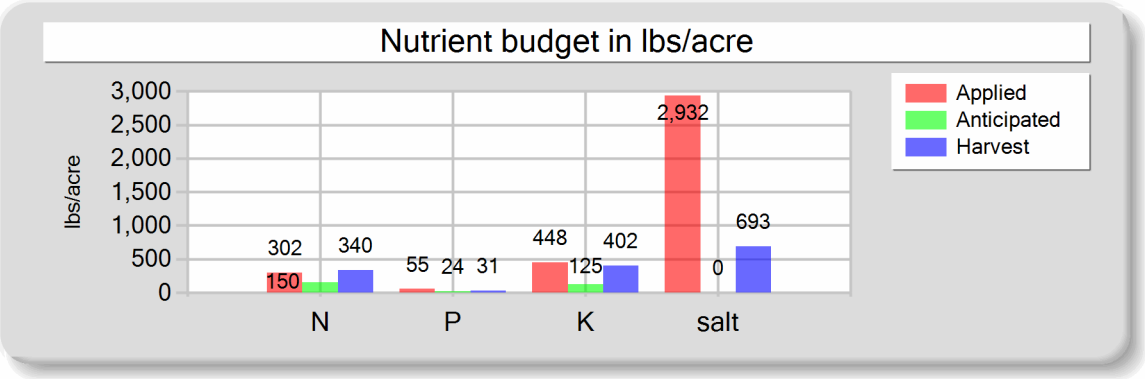
Fresh water applied	
35,544,960.00	*gallons*
1,309.00	*acre-inches*
32.73	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy1 - 09/24/2022: Oats, silage-soft dough

Field name: Dairy1 Crop: Oats, silage-soft dough Plant date: 09/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.51	55.13	448.47	2,898.50
Fresh water	0.00	0.00	0.00	33.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.51	55.13	448.47	2,932.20
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	339.89	31.33	401.96	693.37
Nutrient balance	-38.38	23.80	46.52	2,238.83
Applied to removed ratio	0.89	1.76	1.12	4.23

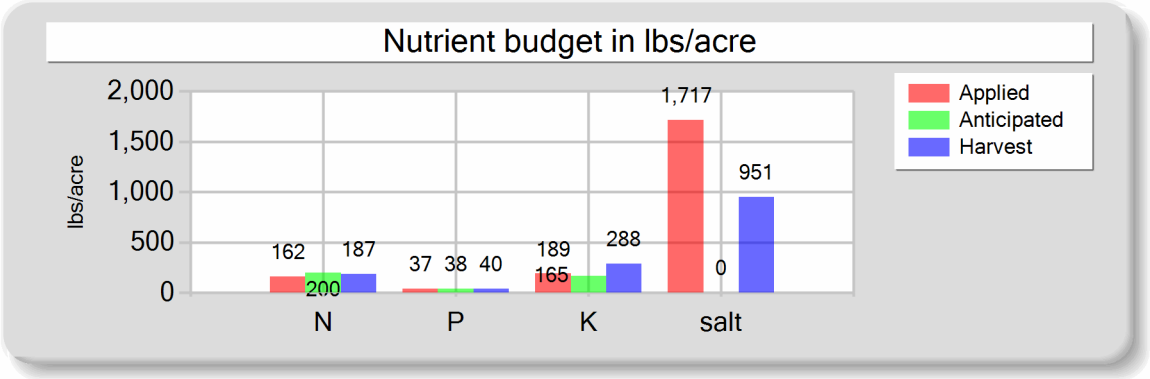
Fresh water applied	
2,692,800.00	*gallons*
99.17	*acre-inches*
5.51	*inches/acre*

Process wastewater applied	
1,530,000.00	*gallons*
56.34	*acre-inches*
3.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy1 - 05/27/2023: Corn, silage

Field name: Dairy1 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.67	36.95	188.99	1,514.34
Fresh water	0.00	0.00	0.00	202.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	161.67	36.95	188.99	1,716.58
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	187.43	40.05	288.36	951.46
Nutrient balance	-25.76	-3.10	-99.37	765.12
Applied to removed ratio	0.86	0.92	0.66	1.80

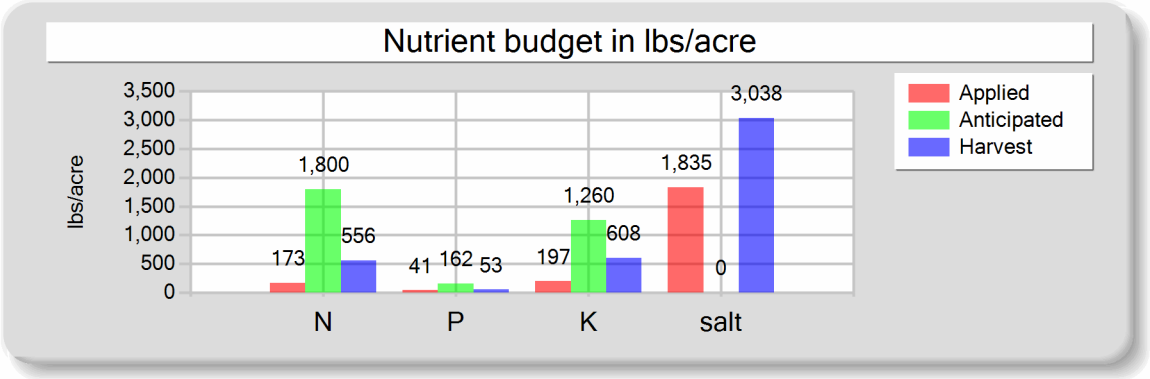
Fresh water applied	
16,156,800.00	*gallons*
595.00	*acre-inches*
33.06	*inches/acre*

Process wastewater applied	
1,110,000.00	*gallons*
40.88	*acre-inches*
2.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy2 - 03/02/2023: Alfalfa, hay

Field name: Dairy2 Crop: Alfalfa, hay Plant date: 03/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	158.76	40.61	196.69	1,591.87
Fresh water	0.00	0.00	0.00	242.69
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	172.76	40.61	196.69	1,834.55
Anticipated crop nutrient removal	1,800.00	162.00	1,260.00	0.00
Actual crop nutrient removal	556.30	52.56	608.05	3,038.29
Nutrient balance	-383.54	-11.94	-411.35	-1,203.73
Applied to removed ratio	0.31	0.77	0.32	0.60

Fresh water applied	
15,079,360.00	*gallons*
555.32	*acre-inches*
39.67	*inches/acre*

Process wastewater applied	
915,000.00	*gallons*
33.70	*acre-inches*
2.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/25/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	2,660.00	7,420.00							28.30
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/25/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	9,050.00	40,800.00							28.80
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/16/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	473.00	277.00	0.00	0.00	90.40	813.00								6,010.00	4,880
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 03/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	339.00	168.00	0.00	0.00	61.00	394.00								3,880.00	3,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 10/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	260.00	165.00	0.00	0.00	83.20	339.00								4,000.00	2,840
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 12/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	272.00	132.00	0.00	0.00	54.60	324.00								3,340.00	2,840
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW 3

IW 3

Fresh Water

Sample description: Fresh Water

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								35.90	27
DL	0.50	0.50	0.50								1.00	10

IW 5

Fresh Water

Sample description: Fresh Water

Sample date: 12/04/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	27.70	0.00	16.50								862.00	465
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

Blaker

Soils

Sample and source description: Soils

Sample date: 12/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			4.00				

Corner1

Soils

Sample and source description: Soils

Sample date: 12/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			4.00				

Dairy1

Soils

Sample and source description: Soils

Sample date: 12/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			58.00				
DL			4.00				

Dairy2

Soils

Sample and source description: Soils

Sample date: 12/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			39.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Blaker - 10/31/2022: Oats, silage-soft dough

Blaker - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	1,870.00	18,700.00		9.71
DL	100.00	100.00	100.00		0.01

Blaker - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,310.00	1,330.00	5,490.00		9.67
DL	100.00	100.00	100.00		0.01

Corner1 - 10/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,840.00	1,230.00	10,500.00		9.24
DL	100.00	100.00	100.00		0.01

Corner1 - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,950.00	910.00	3,770.00		5.84
DL	100.00	100.00	100.00		0.01

Dairy1 - 09/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	1,060.00	13,600.00		10.20
DL	100.00	100.00	100.00		0.01

Dairy1 - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,510.00	750.00	5,400.00		6.96
DL	100.00	100.00	100.00		0.01

Dairy2 - 03/02/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,880.00	650.00	7,520.00		15.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

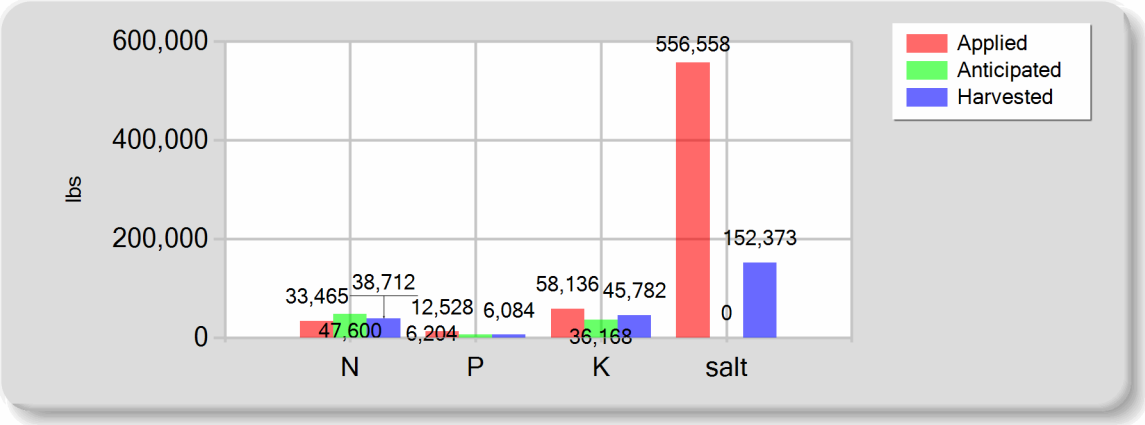
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

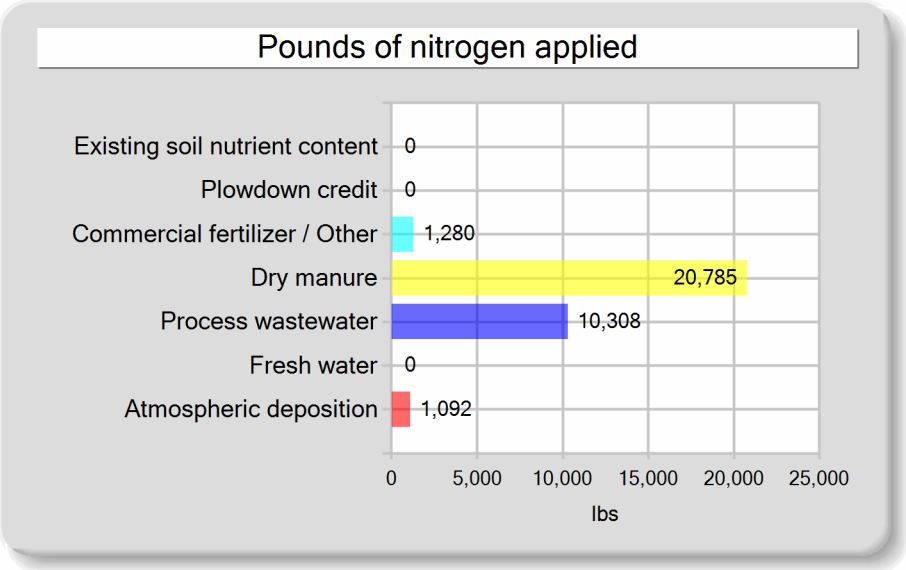
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

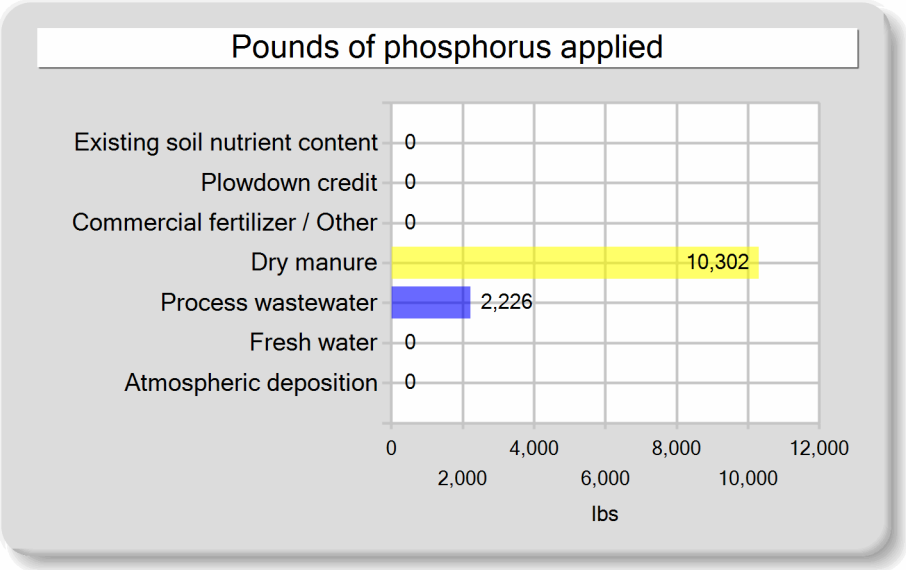
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1,280.00	0.00	0.00	0.00
Dry manure	20,784.68	10,302.34	43,907.60	437,973.54
Process wastewater	10,307.91	2,225.96	14,228.02	101,717.21
Fresh water	0.00	0.00	0.00	16,866.97
Atmospheric deposition	1,092.00	0.00	0.00	0.00
Total nutrients applied	33,464.59	12,528.30	58,135.61	556,557.72
Anticipated crop nutrient removal	47,600.00	6,204.00	36,168.00	0.00
Actual crop nutrient removal	38,712.07	6,083.55	45,782.22	152,372.73
Nutrient balance	-5,247.48	6,444.75	12,353.39	404,184.99
Applied to removed ratio	0.86	2.06	1.27	3.65

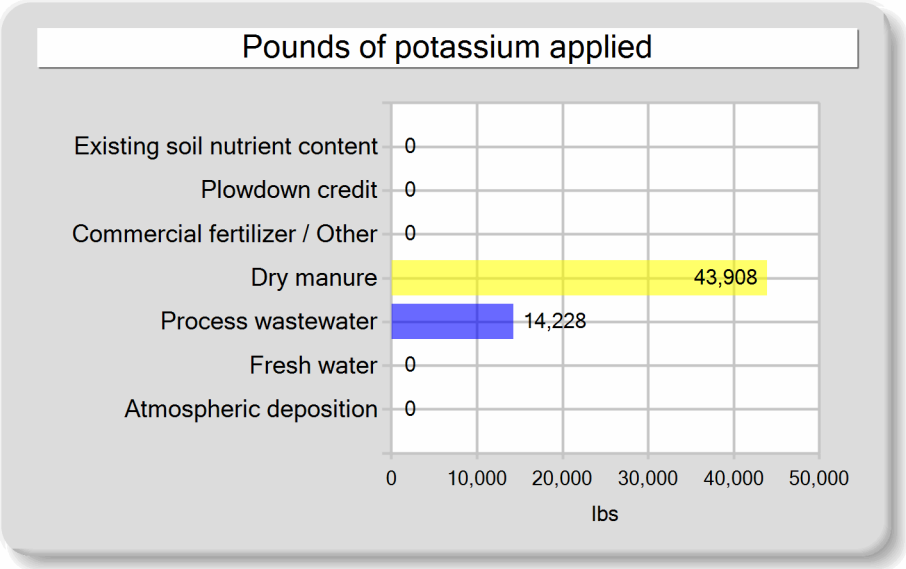
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

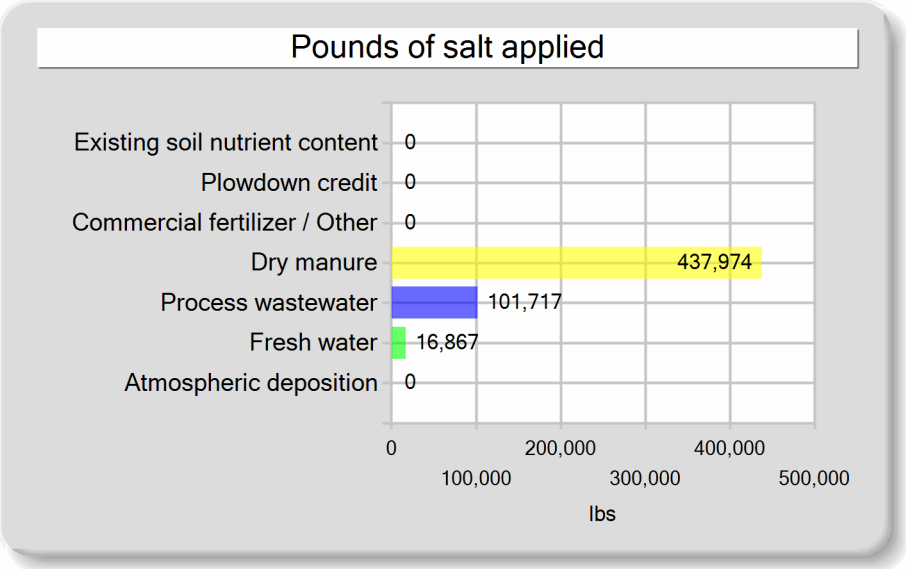


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

andy germann
andy germann (Jun 21, 2024 12:38 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Andy or Charlie Germann	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 21, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Germann Bros
Mailing address: 1236 E. Taylor Rd
(not facility address) Ceres
Phone#:

Authorized Copy Released To: F & R

Sampled By: Jackie Germann
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	10/23/23	11:30	30 WW		12.0°c			✓			
2	10/24/23	11:00	DW 1		9.6°c		✓				
3	10/24/23	11:05	DW 2		9.6°c		✓				
4	10/24/23	11:10	DW 3		9.0°c		✓				
5	10/24/23	10:00	DW 4		10.0°c		✓				
6	10/24/23	9:00	DW 5136		8.8°c		✓				
7	10/23/23	11:00	Corral Solids Spring						✓		ASH
8	10/23/23	11:00	Alfalfa Dairy 2							✓	
9	10/24/23	11:00	Corral Solids Fall								✓
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st						
2nd		DENELE ANALYTICAL, INC.	10-25-23 10:00			T3298003

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Germann Bros. Dairy
1236 E. Taylor Rd.
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/25/23 10:00 AM
SUBMITTED BY: Jackie
ANALYZED DATE: 10/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 9.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T3298003B	Electrical Conductivity (EC)	EPA 120.1 and	0.530	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.59	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	339	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.8	101 (T3298003B)	97.6 (T3298003B)	3.02
Electrical Conductivity (EC)	96.4			0.904
Nitrate Nitrogen (NO3-N)	99.8	103 (T3297053A)	97.9 (T3297053A)	4.92

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Ceres, CA 95307

Grower

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/25/23 10:00 AM
SUBMITTED BY: Jackie
ANALYZED DATE: 10/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 9.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T3298003C	Electrical Conductivity (EC)	EPA 120.1 and	0.941	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.9	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	602	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.8	101 (T3298003B)	97.6 (T3298003B)	3.02
Electrical Conductivity (EC)	96.4			0.904
Nitrate Nitrogen (NO3-N)	99.8	103 (T3297053A)	97.9 (T3297053A)	4.92

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Ceres, CA 95307

Grower

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F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/25/23 10:00 AM
SUBMITTED BY: Jackie
ANALYZED DATE: 10/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 9.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 3	T3298003D	Electrical Conductivity (EC)	EPA 120.1 and	0.961	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	33.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	615	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.8	101 (T3298003B)	97.6 (T3298003B)	3.02
Electrical Conductivity (EC)	96.4			0.904
Nitrate Nitrogen (NO3-N)	99.8	103 (T3297053A)	97.9 (T3297053A)	4.92

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Grower

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F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/25/23 10:00 AM
SUBMITTED BY: Jackie
ANALYZED DATE: 10/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 4	T3298003E	Electrical Conductivity (EC)	EPA 120.1 and	1.58	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	52.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1010	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.8	101 (T3298003B)	97.6 (T3298003B)	3.02
Electrical Conductivity (EC)	96.4			0.904
Nitrate Nitrogen (NO3-N)	99.8	103 (T3297053A)	97.9 (T3297053A)	4.92

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Ceres, CA 95307

Grower

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F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/25/23 10:00 AM
SUBMITTED BY: Jackie
ANALYZED DATE: 10/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 8.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 5136	T3298003F	Electrical Conductivity (EC)	EPA 120.1 and	0.662	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	15.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	424	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.8	101 (T3298003B)	97.6 (T3298003B)	3.02
Electrical Conductivity (EC)	96.4			0.904
Nitrate Nitrogen (NO3-N)	99.8	103 (T3297053A)	97.9 (T3297053A)	4.92

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Germann Bros 2023 Annual Report

Final Audit Report 2024-06-21

Created:	2024-06-19
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAXzeIXAMBpIBghGDtb5CmIqkXGW-k17Lc

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Document created by Vince Furtado (vfurtado@fragservices.com)
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Signer germannjackie@gmail.com entered name at signing as andy germann
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